



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	CLASSICAL MECHANICS				
Type of Course	Core in Major				
Semester	VII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Strong foundation in introductory physics covering Kinematics, Dynamics and basic calculus, alongside a familiarity with vectors, Newton's laws of motion, and mathematical techniques such as differential and integral calculus.				
Course Summary	Exploring topics such as Lagrangian and Hamiltonian Mechanics, Variational principles and coupled oscillations, often incorporating advanced mathematical techniques like differential geometry and calculus of variations.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the principles of calculus of variations and its applications in finding extremals of functionals	Understanding	Conceptual Knowledge	Written exams, quizzes

CO2	Apply variational calculus techniques to solve problems involving optimization and constraint satisfaction	Applying	Procedural Knowledge	Problem sets, simulations
CO3	Analyze the Lagrangian formulation of classical mechanics and its equivalence to Newtonian mechanics	Analyzing	Conceptual Knowledge	Homework assignments, exams
CO4	Derive and interpret the Euler-Lagrange equation and its solutions for various physical systems	Analyzing	Procedural Knowledge	Class discussions, presentations
CO5	Formulate and solve Hamilton's equations of motion for dynamical systems in phase space	Applying	Procedural Knowledge	Laboratory experiments, projects
CO6	Investigate the behavior of coupled oscillators and their dynamics using analytical and numerical methods	Analyzing	Conceptual Knowledge	Research papers, presentations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	CALCULUS OF VARIATIONS		7	12
	1	Introduction; Statement of the Problem	1	
	2	Euler Equations	2	
	3	The 'Second Form' of the Euler Equation	1	
	4	Functions with Several Dependent Variables; Euler Equations when Auxiliary Conditions are Imposed	2	

	5	The Delta Notation	1	
	Sections 6.1 – 6.7 of chapter 6 of Book 1			
II	LAGRANGIAN DYNAMICS		13	20
	6	Introduction; Hamilton's Principle	2	
	7	Generalized Coordinates	1	
	8	Lagrange's Equations of Motion in Generalized Coordinates	3	
	9	Lagrange's Equations with Undetermined Multipliers	2	
	10	Equivalence of Lagrange's and Newton's Equations	1	
	11	Essence of Lagrangian Dynamics	1	
	12	Conservation Theorems	3	
	Sections 7.1 – 7.9 of chapter 7 of Book 1			
III	HAMILTONIAN DYNAMICS		17	25
	13	Canonical Equations of Motion in Hamiltonian Dynamics	3	
	14	Dynamical Variables and Variational Calculus	2	
	15	Phase space and Liouville's Theorem	2	
	16	Virial Theorem	2	
	17	Canonical Transformations	2	
	18	Discovering Three New Forms of the Generating Function	2	
	19	Poisson Brackets	1	
	20	Hamilton – Jacobi Equation	3	
	Sections 7.10 – 7.13 of chapter 7 of Book 1; sections 6.1 – 6.4 of chapter 6 of Book 2			

IV	COUPLED OSCILLATIONS		8	13
	21	Introduction; Two Coupled Harmonic Oscillators; Weak Coupling	3	
	22	General Problem of Coupled Oscillations	2	
	23	Normal Coordinates	2	
	24	Molecular Vibrations	1	
	Sections 12.1 – 12.4, 12.6, 12.7 of chapter 12 of Book 1			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Mode constants of a vibrating strip. To determine the first and second mode constants of a steel vibrating strip; Y to be measured by the Cantilever method and frequency of vibration by the Melde's string method.		
	2	Simulate the orbits for various total energy in central force motion.		
	3	Simulate and verify Rutherford's scattering formula.		
	4	Simulate the Van der Pol oscillator and obtain the limit cycle.		
	5	Simulate and plot the phase space trajectory of a projectile.		
	6	Simulate and plot the phase space trajectory of a simple pendulum in and around the separatrix.		
	7	Simulate the two dimensional harmonic oscillation motion for various phase angles. By tuning the conditions obtain various Lissajous curves. (See section 3.3 of Book 1)		
	8	Simulate the motion (time dependence of position, velocity, energy, rate of loss of energy, etc.) of the damped harmonic oscillator (See section 3.5 of Book 1)		
	9	Simulate the response of linear oscillators to impulsive forcing functions (See section 3.9 of Book 1)		

Books and References:

1. Classical Dynamics of Particles and Systems by Stephen T Thornton and Jerry B. Marion, Fifth edition (Book 1)
2. Analytical Mechanics by Louis N Hand and Janet D Finch (Book 2)
3. A Student's Guide to Lagrangians and Hamiltonians by Patrick Hamill
4. A Student's Guide to Analytical Mechanics by John L Bohn
5. Classical Mechanics by N C Rana and P S Joag
6. Classical Mechanics by Herbert Goldstein, Charles P. Poole Jr. and John L. Safko
7. Classical Mechanics by John R. Taylor
8. Introduction to Classical Mechanics: With Problems and Solutions by David Morin
9. Classical Mechanics: Point Particles and Relativity by Walter Greiner
10. NPTEL video lectures: <https://nptel.ac.in/courses/122106027>
11. Python for Education by Dr. B P Ajithkumar, IUAC, New Delhi; e-book freely downloadable from <https://scischool.in/python/index.html>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PS O5	PS O6	PO 1	PO 2	PO 3	PO 4	PO5	PO6	PO7
CO 1	2	0	1	0	2	1	1	0	1	1	1	1	0
CO 2	2	1	3	1	2	1	1	0	1	1	1	1	0
CO 3	2	0	3	0	2	1	1	0	1	1	1	1	0
CO 4	2	0	3	0	2	1	1	0	1	1	1	1	0
CO 5	2	1	3	1	2	1	1	0	1	1	1	1	0
CO 6	2	2	3	1	2	2	1	0	1	1	2	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	